Week 6 Report

Computational Statistics for Data Science · ISS507700

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112006211 helped in discussing for qualitative assessment of 2(e)

I have largely solved the problems without the use of any AI agent or tool.

Q0 · Setup

```
# read each media dataset (1-4 correspond to the 4 media types)
m1 <- read.csv("pls-media/pls-media1.csv")
m2 <- read.csv("pls-media/pls-media2.csv")
m3 <- read.csv("pls-media/pls-media3.csv")
m4 <- read.csv("pls-media/pls-media4.csv")

# combine into one dataset with a group label
media_data <- rbind(
  data.frame(media = "1_video_animation_audio", INTEND.0 = m1$INTEND.0),
  data.frame(media = "2_video_pictures_audio",  INTEND.0 = m2$INTEND.0),
  data.frame(media = "3_webpage_pictures_text", INTEND.0 = m3$INTEND.0),
  data.frame(media = "4_webpage_text_only",     INTEND.0 = m4$INTEND.0)
)

# remove missing values (if any)
media_data <- na.omit(media_data)
```

Q1 · Explore and Describe the Data.

(a) The means of viewers' intentions to share (INTEND.0) on each of the four media types.

```r
groups <- split(media_data$INTEND.0, media_data$media)
group_means <- sapply(groups, mean)
group_means   # shows average intention per media
```

```
#> 1_video_animation_audio  2_video_pictures_audio 3_webpage_pictures_text
#>                4.809524                3.947368                4.725000
#>     4_webpage_text_only
#>                4.891304
```

I

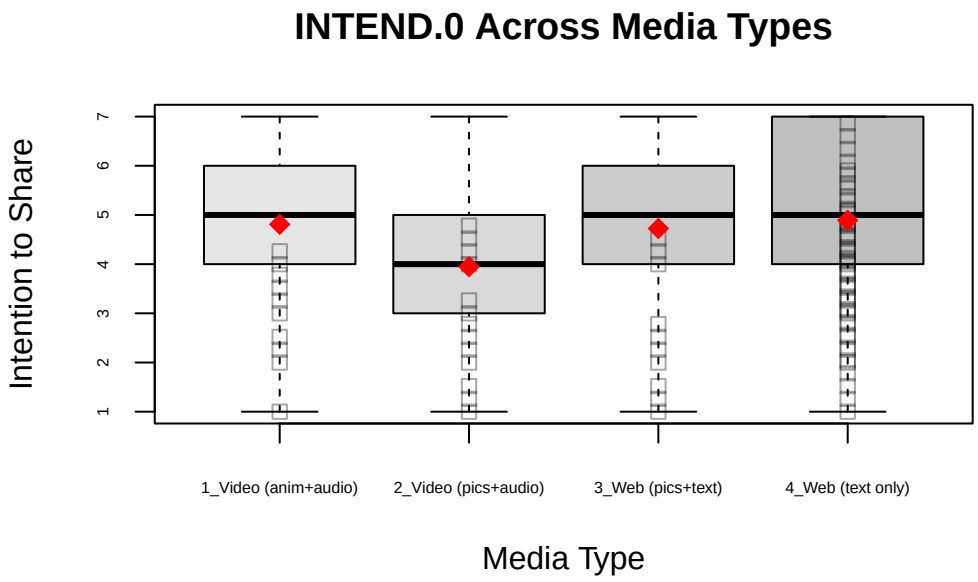
(b) Visualize the distribution and mean of intention to share, across all four media.

```r
media_data$media <- factor(media_data$media,
  labels = c("1_Video (anim+audio)",
             "2_Video (pics+audio)",
             "3_Web (pics+text)",
             "4_Web (text only)"))

# boxplot shows spread and median for each group
boxplot(INTEND.0 ~ media, data = media_data,
        col = c("gray90","gray85","gray80","gray75"),
        main = "INTEND.0 Across Media Types",
        xlab = "Media Type",
        ylab = "Intention to Share",
        cex.axis = 0.5)

# add individual data points (to see distribution shape)
stripchart(INTEND.0 ~ media, data = media_data,
           method = "stack",
           add = TRUE, col = rgb(0,0,0,0.35))

# add group means as red points
points(1:4, group_means, col = "red", pch = 18, cex = 1.5)
```



(c) From the visualization alone, does the media type make a difference on intention to share?

There is a slight difference between media types, but it is not clearly strong.

The second media type (video with pictures and audio) has a lower average intention to share, while the others are slightly higher and quite similar to each other. However, the distributions of all groups overlap heavily, and each group shows a wide range of responses. This means the differences between groups are small compared to the variability within groups, so the effect of media type is not clearly strong based on the visualization alone.

Q2 · Traditional One-way ANOVA.

(a) The null and alternative hypotheses when comparing INTEND.0 across four groups in ANOVA.

H_0 (null): The mean intention to share is the same across all four media types.
H_1 (alternative): At least one media type has a different mean intention to share.

(b) Compute the F-statistic.

(i) Code and results of computing MSTR, MSE, and F.

```
# split data into groups
groups <- split(media_data$INTEND.0, media_data$media)

# get group sizes, means, variances
n_j <- sapply(groups, length)
xbar_j <- sapply(groups, mean)
s2_j <- sapply(groups, var)

# total groups and sample size
k <- length(groups)
N <- sum(n_j)

# overall mean (across all data)
grand_mean <- mean(media_data$INTEND.0)

# between-group variation (SSTR)
SSTR <- sum(n_j * (xbar_j - grand_mean)^2)

# within-group variation (SSE)
SSE  <- sum((n_j - 1) * s2_j)

# degrees of freedom
df_MSTR <- k - 1
df_MSE  <- N - k

# mean squares
MSTR <- SSTR / df_MSTR   # variation between groups
MSE  <- SSE / df_MSE     # variation within groups

# F statistic = ratio of between vs within variation
F_value <- MSTR / MSE

# print results
MSTR
```

```
#> [1] 7.507617
```

```
MSE
```

```
#> [1] 2.869151
```

```
F_value
```

```
#> [1] 2.616669
```

(ii) Compute the p-value of F, from the null F-distribution; is the F-value significant? If so, state your conclusion for the hypotheses.

```
# p-value from F distribution
p_value_F <- pf(F_value, df1 = df_MSTR, df2 = df_MSE, lower.tail = FALSE)

# print results
p_value_F
```

```
#> [1] 0.05289015
```

> The F value is not significant.

The p-value of the F-statistic is 0.0529. Since this is greater than 0.05, the result is not statistically significant at the 5% level. Therefore, we fail to reject the null hypothesis, and conclude that there is no significant difference in mean intention to share across the four media types.

(c) Conduct the same one-way ANOVA using the aov() function in R – confirm that you got similar results.

```
anova_model <- aov(INTEND.0 ~ factor(media), data = media_data)
summary(anova_model)
```

```
#>                Df Sum Sq Mean Sq F value Pr(>F)
#> factor(media)   3   22.5   7.508   2.617 0.0529 .
#> Residuals     162  464.8   2.869
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> Same with manual F result (2.617 for F and 0.0529 for p-value).

(d) Regardless of your conclusions, conduct a post-hoc Tukey test to see if any pairs of media have significantly different means – what do you find?

```r
TukeyHSD(anova_model, conf.level = 0.95)
```

```
#>   Tukey multiple comparisons of means
#>     95% family-wise confidence level
#>
#> Fit: aov(formula = INTEND.0 ~ factor(media), data = media_data)
#>
#> $`factor(media)`
#>                                                 diff         lwr       upr
#> 2_Video (pics+audio)-1_Video (anim+audio) -0.86215539 -1.84660332 0.1222925
#> 3_Web (pics+text)-1_Video (anim+audio)    -0.08452381 -1.05596494 0.8869173
#> 4_Web (text only)-1_Video (anim+audio)     0.08178054 -0.85664966 1.0202107
#> 3_Web (pics+text)-2_Video (pics+audio)     0.77763158 -0.21843807 1.7737012
#> 4_Web (text only)-2_Video (pics+audio)     0.94393593 -0.01996662 1.9078385
#> 4_Web (text only)-3_Web (pics+text)        0.16630435 -0.78431033 1.1169190
#>                                                    p adj
#> 2_Video (pics+audio)-1_Video (anim+audio) 0.1085727
#> 3_Web (pics+text)-1_Video (anim+audio)    0.9959223
#> 4_Web (text only)-1_Video (anim+audio)    0.9959032
#> 3_Web (pics+text)-2_Video (pics+audio)    0.1825044
#> 4_Web (text only)-2_Video (pics+audio)    0.0573229
#> 4_Web (text only)-3_Web (pics+text)       0.9687417
```

No pair of media types shows a statistically significant difference in mean intention to share.

The Tukey post-hoc test shows that no pair of media types has a statistically significant difference in mean intention to share at the 5% level, since all adjusted p-values are greater than 0.05 and all confidence intervals include 0.

The comparison between media 4 (webpage text only) and media 2 (video with pictures and audio) has the smallest p-value (≈ 0.057) and the largest mean difference, suggesting a potential difference. However, it still does not reach statistical significance at the 5% level.

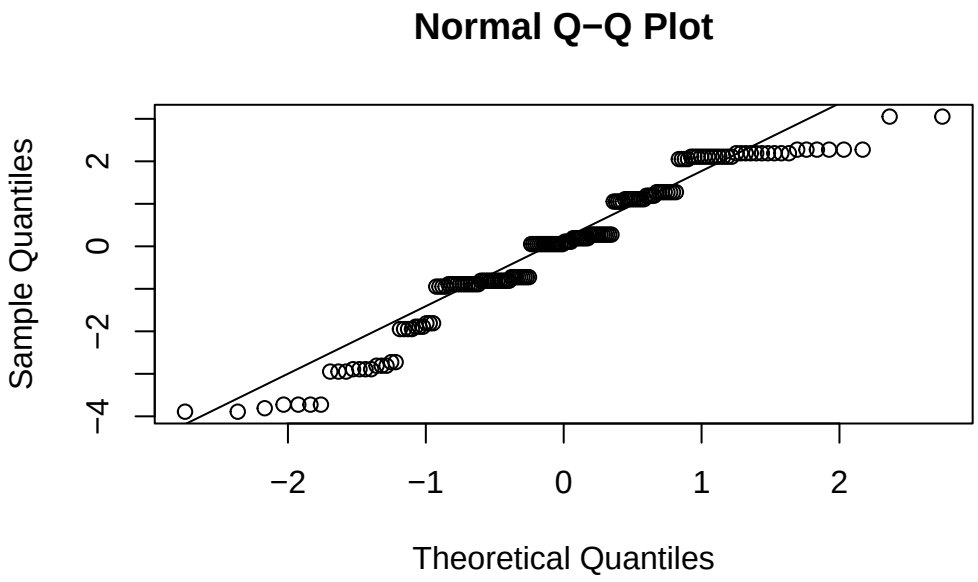
Therefore, we conclude that there is no significant pairwise difference between any media types.

(e) Were the classic requirements of one-way ANOVA met?

The ANOVA assumptions were partially met.

Proof:

```r
# normality check (residuals)
qqnorm(residuals(anova_model))
qqline(residuals(anova_model))
```




```
# Shapiro-Wilk test (per group)
by(media_data$INTEND.0, media_data$media, shapiro.test)
```

```
#> media_data$media: 1_Video (anim+audio)
#>
#>  Shapiro-Wilk normality test
#>
#> data:  dd[x, ]
#> W = 0.91279, p-value = 0.003557
#>
#> ------------------------------------------------------------
#> media_data$media: 2_Video (pics+audio)
#>
#>  Shapiro-Wilk normality test
#>
#> data:  dd[x, ]
#> W = 0.92974, p-value = 0.01969
#>
#> ------------------------------------------------------------
#> media_data$media: 3_Web (pics+text)
#>
#>  Shapiro-Wilk normality test
#>
#> data:  dd[x, ]
#> W = 0.88247, p-value = 0.0006139
#>
#> ------------------------------------------------------------
```

```
#> media_data$media: 4_Web (text only)
#>
#>  Shapiro-Wilk normality test
#>
#> data:  dd[x, ]
#> W = 0.89611, p-value = 0.0006242
```

For normality, Q–Q plots of the residuals show noticeable deviations from the straight line, and Shapiro–Wilk tests on each group yield $p < 0.05$, indicating that normality is violated. This is not surprising since INTEND.0 is measured on a discrete 1–7 scale. However, group sizes are relatively balanced, so ANOVA is somewhat robust to this violation.

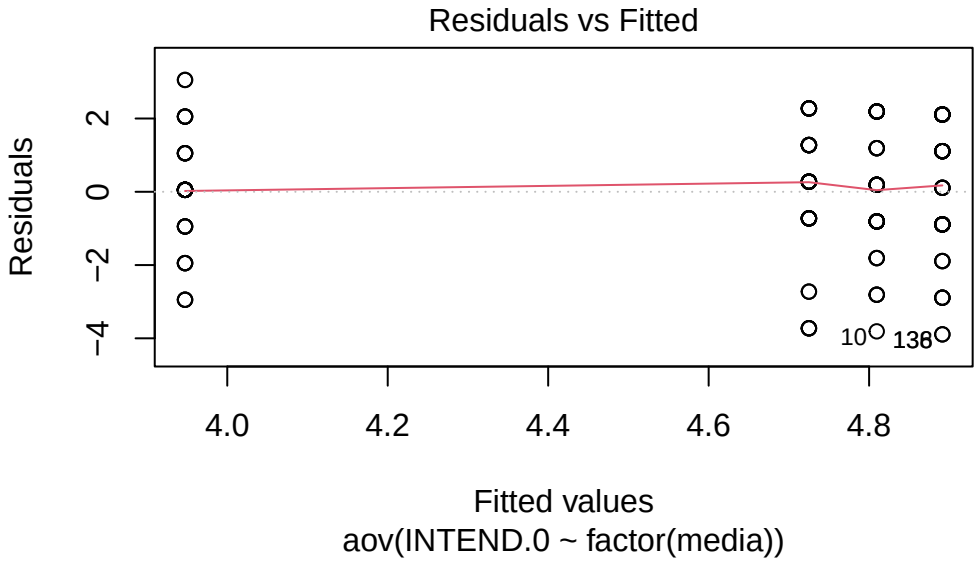
```
# equal variance tests
bartlett.test(INTEND.0 ~ media, data = media_data)
```

```
#>
#>  Bartlett test of homogeneity of variances
#>
#> data:  INTEND.0 by media
#> Bartlett's K-squared = 1.3958, df = 3, p-value = 0.7065
```

```
fligner.test(INTEND.0 ~ media, data = media_data)
```

```
#>
#>  Fligner-Killeen test of homogeneity of variances
#>
#> data:  INTEND.0 by media
#> Fligner-Killeen:med chi-squared = 3.5095, df = 3, p-value = 0.3195
```

```
# residuals vs fitted (variance check)
plot(anova_model, which = 1)
```



For equal variance, the spreads across groups appear similar in the boxplot (see Q1), and the sample variances are close (approximately 2.32, 2.69, 3.08, 3.30). Formal tests such as Bartlett or Fligner are not significant ($p > 0.05$), supporting the assumption of homogeneity of variances.

For independence, this assumption cannot be tested directly, but it is reasonable given the experimental design where participants were randomly assigned to one media condition. Therefore, independence is likely satisfied.

Overall, while ANOVA results are usable, the violation of normality suggests that the Kruskal–Wallis test provides a more reliable conclusion (proven in Q3).

Q3 · Non-parametric Kruskal Wallis Test.

(a) The null and alternative hypotheses.

H_0 (null): The distribution (median) of intention to share is the same across all four media types.
H_1 (alternative): At least one media type has a different distribution (median).

(b) Compute an approximate Kruskal Wallis H ourselves.

(i) Show the code and results of computing H.

```
# rank all values (ignores group labels)
all_ranks <- rank(media_data$INTEND.0)
```

```r
# split ranks back into groups
group_ranks <- split(all_ranks, media_data$media)

# sum of ranks per group
R_i <- sapply(group_ranks, sum)

# group sizes
n_i <- sapply(group_ranks, length)

# compute H statistic
H <- (12 / (N * (N + 1))) * sum((R_i^2) / n_i) - 3 * (N + 1)

# tie correction
tie_tab <- table(media_data$INTEND.0)
C <- 1 - sum(tie_tab^3 - tie_tab) / (N^3 - N)

# corrected H
H_corrected <- H / C

H
```

```
#> [1] 8.45466
```

```r
H_corrected
```

```
#> [1] 8.828334
```

(ii) Compute the p-value of H, from the null chi-square distribution; is the H value significant at 5% significance? If so, state your conclusion of the hypotheses.

```r
# p-value from chi-square distribution
p_value_H <- 1 - pchisq(H, df = k - 1)
p_value_H_corrected <- 1 - pchisq(H_corrected, df = length(n_i) - 1)

# print result
p_value_H
```

```
#> [1] 0.03749292
```

```r
p_value_H_corrected
```

```
#> [1] 0.03166253
```

The H value is significant.

The uncorrected p-value is 0.03749, and the tie-corrected p-value is 0.03166; both are less than 0.05, so we reject H_0.

Therefore, we conclude that at least one media type differs in the distribution (median) of intention to share. Although the corrected value is more accurate due to ties in the data, both versions lead to the same conclusion.

(c) Conduct the same test using the kruskal.test() function in R – confirm that you got similar results.

```
kruskal.test(INTEND.0 ~ media, data = media_data)
```

```
#>
#>  Kruskal-Wallis rank sum test
#>
#> data:  INTEND.0 by media
#> Kruskal-Wallis chi-squared = 8.8283, df = 3, p-value = 0.03166
```

Same result with the tie-corrected manual result (H=8.8283 and p=0.03166).

(d) Regardless of your conclusions, conduct a post-hoc Dunn test to see if the values of any pairs of media are significantly different – what are your conclusions?

```
library("FSA")
dunnTest(INTEND.0 ~ media, data = media_data, method = "bonferroni")
```

```
#>                                    Comparison           Z     P.unadj
#> 1 1_Video (anim+audio) - 2_Video (pics+audio)  2.30087819 0.021398517
#> 2    1_Video (anim+audio) - 3_Web (pics+text) -0.09233644 0.926430736
#> 3    2_Video (pics+audio) - 3_Web (pics+text) -2.36408588 0.018074622
#> 4    1_Video (anim+audio) - 4_Web (text only) -0.31452459 0.753122646
#> 5    2_Video (pics+audio) - 4_Web (text only) -2.65613380 0.007904225
#> 6       3_Web (pics+text) - 4_Web (text only) -0.21613379 0.828883460
#>        P.adj
#> 1 0.12839110
#> 2 1.00000000
#> 3 0.10844773
#> 4 1.00000000
#> 5 0.04742535
#> 6 1.00000000
```

Only media 2 (video with pictures + audio) and media 4 (webpage text only) differ significantly.

The Dunn test shows that only the pair between media 2 (video with pictures + audio) and media 4 (webpage text only) is significantly different after Bonferroni adjustment. This supports the result of the Kruskal–Wallis test in (ii), which indicated that at least one group differs.

Although ANOVA was not significant ($p = 0.0529$), the Kruskal–Wallis test was significant ($p = 0.0317$), suggesting that differences across media types exist, and given the violation of normality, the non-parametric result is more reliable.